B Model Architectures and Sizes

As we showed in section 5, capacity is an important parameter for learning strong cross-lingual representations. In the table below, we list multiple monolingual and multilingual models used by the research community and summarize their architectures and total number of parameters.

Model	#lgs	tokenization	L	H_m	H_{ff}	A	V	#params
$\text{BERT}_{\text{Base}}$	1	WordPiece	12	768	3072	12	30k	110M
$\text{BERT}_{\text{Large}}$	1	WordPiece	24	1024	4096	16	30k	335M
mBERT	104	WordPiece	12	768	3072	12	110k	172M
$\text{RoBERTa}_{\text{Base}}$	1	bBPE	12	768	3072	8	50k	125M
RoBERTa	1	bBPE	24	1024	4096	16	50k	355M
XLM-15	15	BPE	12	1024	4096	8	95k	250M
XLM-17	17	BPE	16	1280	5120	16	200k	570M
XLM-100	100	BPE	16	1280	5120	16	200k	570M
Unicoder	15	BPE	12	1024	4096	8	95k	250M
XLM-R $_{\text{Base}}$	100	SPM	12	768	3072	12	250k	270M
XLM-R	100	SPM	24	1024	4096	16	250k	550M
GPT2	1	bBPE	48	1600	6400	32	50k	1.5B
wide-mmNMT	103	SPM	12	2048	16384	32	64k	3B
deep-mmNMT	103	SPM	24	1024	16384	32	64k	3B
T5-3B	1	WordPiece	24	1024	16384	32	32k	3B
T5-11B	1	WordPiece	24	1024	65536	32	32k	11B

Table 7: **Details on model sizes.** We show the tokenization used by each Transformer model, the number of layers L, the number of hidden states of the model H_m, the dimension of the feed-forward layer H_{ff}, the number of attention heads A, the size of the vocabulary V and the total number of parameters #params. For Transformer encoders, the number of parameters can be approximated by $4LH_m^2 + 2LH_mH_{ff} + VH_m$. GPT2 numbers are from Radford et al. (2019), mm-NMT models are from the work of Arivazhagan et al. (2019) on massively multilingual neural machine translation (mmNMT), and T5 numbers are from Raffel et al. (2019). While *XLM-R* is among the largest models partly due to its large embedding layer, it has a similar number of parameters than XLM-100, and remains significantly smaller that recently introduced Transformer models for multilingual MT and transfer learning. While this table gives more hindsight on the difference of capacity of each model, note it does not highlight other critical differences between the models.